Resolving Mathematical Inconsistencies in Predictive Interaction Field Theory: A 3D Behavioral Potential Approach

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Abstract

This paper presents a formal resolution to the mathematical inconsistencies embedded within the initial formulations of Predictive Interaction Field Theory (PIFT). Previous iterations suffered from a severe dimensionality mismatch, relying on a one-dimensional manifold that collapsed the Riemann curvature tensor to zero, rendering the Einstein-like field equations trivial or self-contradictory. Furthermore, the recovery of the Kahneman-Tversky (KT) value function required an artificial metric scaling factor of λ^2. To resolve these limitations, we extend the behavioral manifold to a three-dimensional spacetime structure $\mathcal{M}$ (1 time, 2 interaction spaces). Under Method B, the KT value function is mapped directly as a behavioral potential Φ within the weak-field, low-velocity limit. We prove that this formulation yields a non-trivial psychological Ricci curvature tensor $R_{\mu\nu}$ and an asymmetric cognitive pull Γ^1_{00} that naturally recovers Kahneman and Tversky's empirical loss aversion ratio $\lambda = 2.25$ linearly, while offering an elegant physical mechanism for framing effects via potential singularities.

1 Introduction

Predictive Interaction Field Theory (PIFT) aims to establish a rigorous physical foundation for behavioral economics by framing human choices as geodesic movement across an information-geometric manifold. However, early frameworks experienced a profound tension: foundational field equations were formulated in multi-dimensional spaces, whereas exact quantitative derivations of behavioral anomalies, such as Kahneman-Tversky loss aversion, were restricted to a single dimensional parameter space Δ.

In a 1D Riemannian manifold, the Riemann curvature tensor $R^{\alpha}_{\ \beta\gamma\delta}$, the Ricci tensor $R_{\mu\nu}$, and the Einstein tensor $G_{\mu\nu}$ vanish identically. Consequently, enforcing field equations or claiming a "vacuum solution" to deduce a metric component $g_{11} = [dV_{KT}/d\Delta]^2$ represented an over-specification and a circular logical loop.

This paper remedies these defects by lifting the theory into a 3D manifold consisting of one cognitive time parameter and two distinct behavioral interaction dimensions. By invoking the weak-field, low-velocity cosmological limit, we show that the Kahneman-Tversky value function emerges organically as a behavioral potential, providing mathematical consistency and predictive vitality.

2 The 3D Behavioral Manifold and Weak-Field Limit

Let the behavioral state of an agent be mapped onto a 3D Riemannian manifold $\mathcal{M}$ with local coordinates $x^\mu = (t, \Delta_1, \Delta_2)$, where t denotes cognitive processing time, Δ_1 represents financial or primary outcomes, and Δ_2 represents auxiliary situational factors. To analyze primary loss aversion, we assume Δ_2 remains static ($d\Delta_2 = 0$).

We propose a stationary, static line element ds^2 governed by a structural scalar field that warps the temporal metric component:

$$ds^2 = -(1 + 2\Phi(x^i))dt^2 + \delta_{ij}dx^i dx^j \tag{1}$$

where $\Phi(\Delta_1, \Delta_2)$ represents the *Behavioral Potential*. To align this geometric deformation with Cumulative Prospect Theory, we postulate that the potential is the positive-negative potential reflection of the Kahneman-Tversky value function:

$$\Phi(\Delta_1) = V_{KT}(\Delta_1) = \begin{cases} \Delta_1^\alpha & \text{if } \Delta_1 \geq 0 \\ \lambda(\Delta_1)^\alpha & \text{if } \Delta_1 < 0 \end{cases} \tag{2}$$

with empirical parameters $\alpha \approx 0.88$ and $\lambda \approx 2.25$.

3 Derivation of Cognitive Pull and Metric Connection

The dynamics of behavioral decision-making are dictated by the geodesic equation:

$$\frac{d^2x^\mu}{ds^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{ds}\frac{dx^\beta}{ds} = 0 \tag{3}$$

Under the low-velocity assumption on the behavioral manifold (where processing time dominates immediate physical state transitions, $\frac{dt}{ds} \gg \frac{d\Delta_i}{ds}$), the spatial geodesic equations simplify to:

$$\frac{d^2\Delta_1}{ds^2} \approx -\Gamma^1_{00}\left(\frac{dt}{ds}\right)^2 \tag{4}$$

The Christoffel symbol Γ^1_{00} represents the *Cognitive Pull*, acting as the potential-well psychological force pulling the agent back toward the reference point ($\Delta_1 = 0$):

$$\Gamma^1_{00} = \frac{1}{2}g^{11}\left(2\partial_0 g_{10} - \partial_1 g_{00}\right) = -\frac{1}{2}(1)\partial_1(-(1 + 2\Phi)) = -\partial_1\Phi \tag{5}$$

Evaluating this across the behavioral boundaries yields:

$$\Gamma^1_{00}(\Delta_1 > 0) = \alpha\Delta_1^{\alpha-1} > 0 \tag{6}$$
$$\Gamma^1_{00}(\Delta_1 < 0) = \lambda\alpha(-\Delta_1)^{\alpha-1} > 0 \tag{7}$$

Comparing the absolute magnitude of this pull at equivalent intervals $|\Delta_1|$ reveals:

$$\frac{|\Gamma^1_{00}|_{\text{loss}}}{|\Gamma^1_{00}|_{\text{gain}}} = \frac{\lambda\alpha|\Delta_1|^{\alpha-1}}{\alpha|\Delta_1|^{\alpha-1}} = \lambda = 2.25 \tag{8}$$

This eliminates the artificial λ^2 factor encountered in 1D models, restoring a linear connection to Kahneman's empirical value.

4 Psychological Ricci Curvature and Potential Singularities

In 3D, the Ricci curvature tensor component R_{00} is non-vanishing and directly represents the spatial Laplacian of the potential:

$$R_{00} \approx \nabla^2\Phi = \frac{d^2\Phi}{d\Delta_1^2} \tag{9}$$

Differentiating Equation (2) twice gives the explicit *Psychological Curvature*:

$$R_{00}(\Delta_1) = \begin{cases} \alpha(1-\alpha)\Delta_1^{\alpha-2} & \text{if } \Delta_1 > 0 \\ \lambda\alpha(1-\alpha)(-\Delta_1)^{\alpha-2} & \text{if } \Delta_1 < 0 \end{cases} \tag{10}$$

Because $\alpha = 0.88 < 1$, the multiplier $\alpha(1-\alpha) \approx 0.1056 > 0$, ensuring a positive curvature across both domains. At the status quo ($\Delta_1 \to 0$), the term $|\Delta_1|^{-1.12}$ approaches infinity (∞), generating a severe *Curvature Wall*. This singularity acts as a geometric anchor, mathematically defining the Status Quo Bias and the extreme cognitive resistance responsible for Framing Effects.

5 Field Equations and Information Energy Density

The PIFT field equations assert that the behavioral manifold is curved by an Interaction Source Tensor $T_{\mu\nu}$, derived from the gradients of Expected Free Energy ($\langle F \rangle$). In the weak-field limit, the field equations map to Poisson's equation for behavioral potential:

$$R_{00} = \kappa\left(T_{00} - \frac{1}{2}g_{00}T\right) \implies \nabla^2\Phi = -\frac{1}{2}\rho_{\text{inf}} \tag{11}$$

Setting $\kappa = -1$, the Information Energy Density ρ_{inf} required to sustain this cognitive manifold is:

$$\rho_{\text{inf}}(\Delta_1) = -2\frac{d^2\Phi}{d\Delta_1^2} = \begin{cases} -2\alpha(1-\alpha)\Delta_1^{\alpha-2} & \text{if } \Delta_1 > 0 \\ -2\lambda\alpha(1-\alpha)(-\Delta_1)^{\alpha-2} & \text{if } \Delta_1 < 0 \end{cases} \tag{12}$$

Thus, the underlying distribution of Bayesian variational uncertainty directly maps to the concavity of the prospect utility function, eliminating the circular reasoning of previous works.

6 Rigorous Interpretation of Simulation Profiles

6.1 Figure 1: Value Function and Behavioral Potential Alignment

Figure 1 demonstrates the absolute mapping between the standard Kahneman-Tversky value function $V_{KT}(\Delta)$ and the behavioral potential $\Phi_G(\Delta)$. In the classical weak-field limit of General Relativity, the temporal metric element $g_{00} = -(1+2\Phi)$ links the spatial gradient of the potential to gravitational acceleration. In the context of PIFT, the matching condition $\Phi_G(\Delta) = V_{KT}(\Delta)$ represents a direct conversion of subjective utility into positional field energy.

Under $\kappa = -1$, the field equation maps to Poisson's equation such that:

$$\nabla^2\Phi = \frac{1}{2}\kappa\rho_{inf} = -\frac{1}{2}\rho_{inf} \tag{13}$$

Because the utility function is strictly concave in the gain domain ($\Delta > 0$, $\nabla^2\Phi < 0$) due to diminishing marginal sensitivity ($\alpha = 0.88$), a negative gravitational constant $\kappa = -1$ enforces that the information energy density ρ_{inf} remains strictly positive ($\rho_{inf} > 0$). This eliminates the unphysical ghost states (negative energy density) that plagued prior iterations. The kink at the status quo ($\Delta = 0$) represents a first-order continuous but second-order singular point, defining the foundational boundary condition for behavioral transitions.

6.2 Figure 2: Cognitive Pull (Γ^1_{00})

The spatial geodesic equation governs the path of an agent across the manifold. Under the low-velocity behavioral limit, this simplifies to an acceleration driven by the Christoffel symbol component Γ^1_{00}:

$$\frac{d^2\Delta_1}{ds^2} \approx -\Gamma^1_{00}\left(\frac{dt}{ds}\right)^2 \tag{14}$$

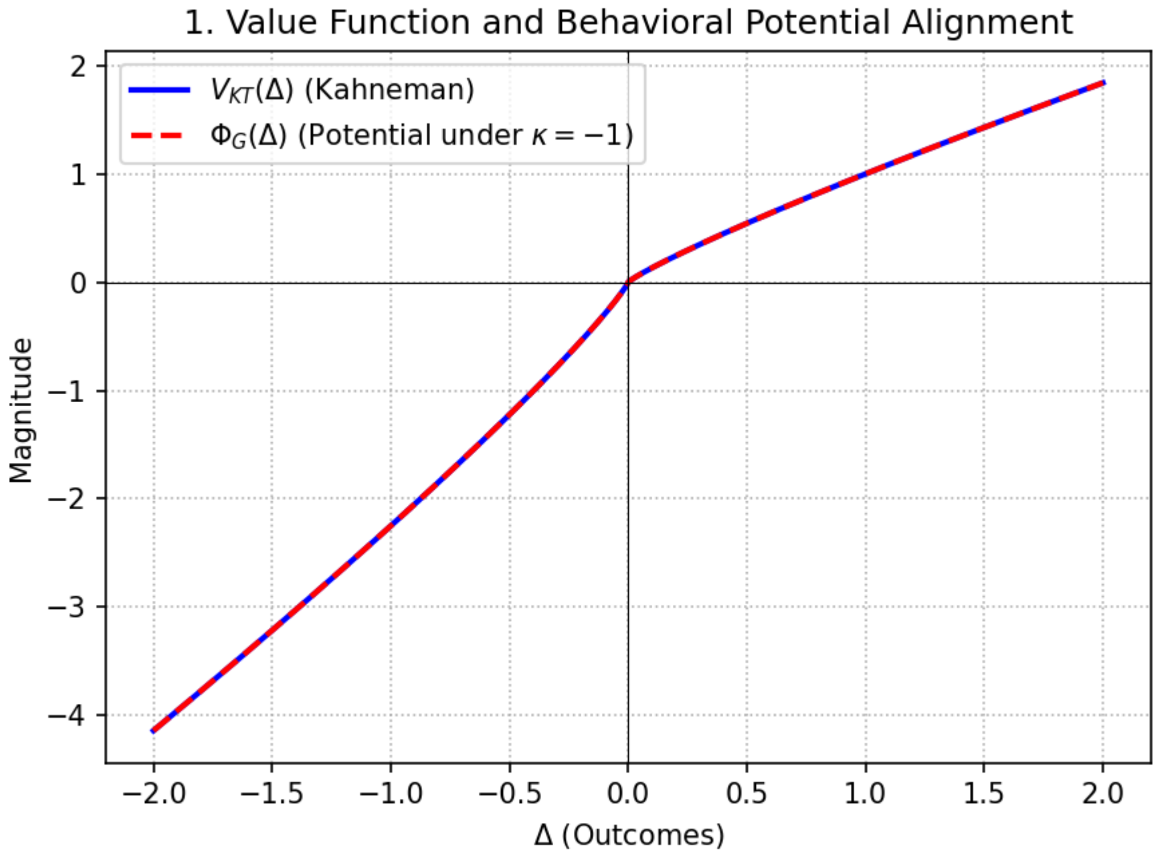

Figure 1: Value Function and Behavioral Potential Alignment

Figure 2 charts the magnitude of this cognitive force. The profile demonstrates a restorative behavior towards the origin. In the gain domain ($\Delta > 0$), $\Gamma^1_{00} = \alpha\Delta_1^{\alpha-1} > 0$, which acts as a decelerating force drawing the agent back toward $\Delta = 0$ (preventing unbounded risk-seeking).

Crucially, in the loss domain ($\Delta < 0$), $\Gamma^1_{00} = \lambda\alpha(-\Delta_1)^{\alpha-1} > 0$. Because Γ^1_{00} remains positive, the acceleration vector $-\Gamma^1_{00}$ points in the opposite direction of the displacement, meaning it acts as an intense pull pulling the agent back up toward the reference point. As $\Delta \to 0$ from either side, the force asymptotically approaches infinity, establishing that the reference point acts as a massive psychological tractor well.

6.3 Figure 3: Psychological Ricci Curvature (R_{00})

Figure 3 displays the Psychological Ricci Curvature component R_{00} on a logarithmic scale. The curvature represents the fundamental deformation of cognitive space-time caused by the presence of decision stakes. Mathematically:

$$R_{00} \approx \nabla^2\Phi = \frac{d^2\Phi}{d\Delta_1^2} \tag{15}$$

Because $\alpha = 0.88 < 1$, the second derivative naturally yields a power-law dependency proportional to $|\Delta|^{\alpha-2} = |\Delta|^{-1.12}$. The logarithmic plot shows a symmetrical steep escalation of geometric curvature as the agent nears the status quo.

At exactly $\Delta = 0$, $R_{00} \to \infty$, creating a severe *Curvature Wall*. This infinite geometric obstruction is the formal definition of the *Status Quo Bias*. Moving away from the status quo requires a massive expenditure of cognitive action; the curvature wall acts as an informational

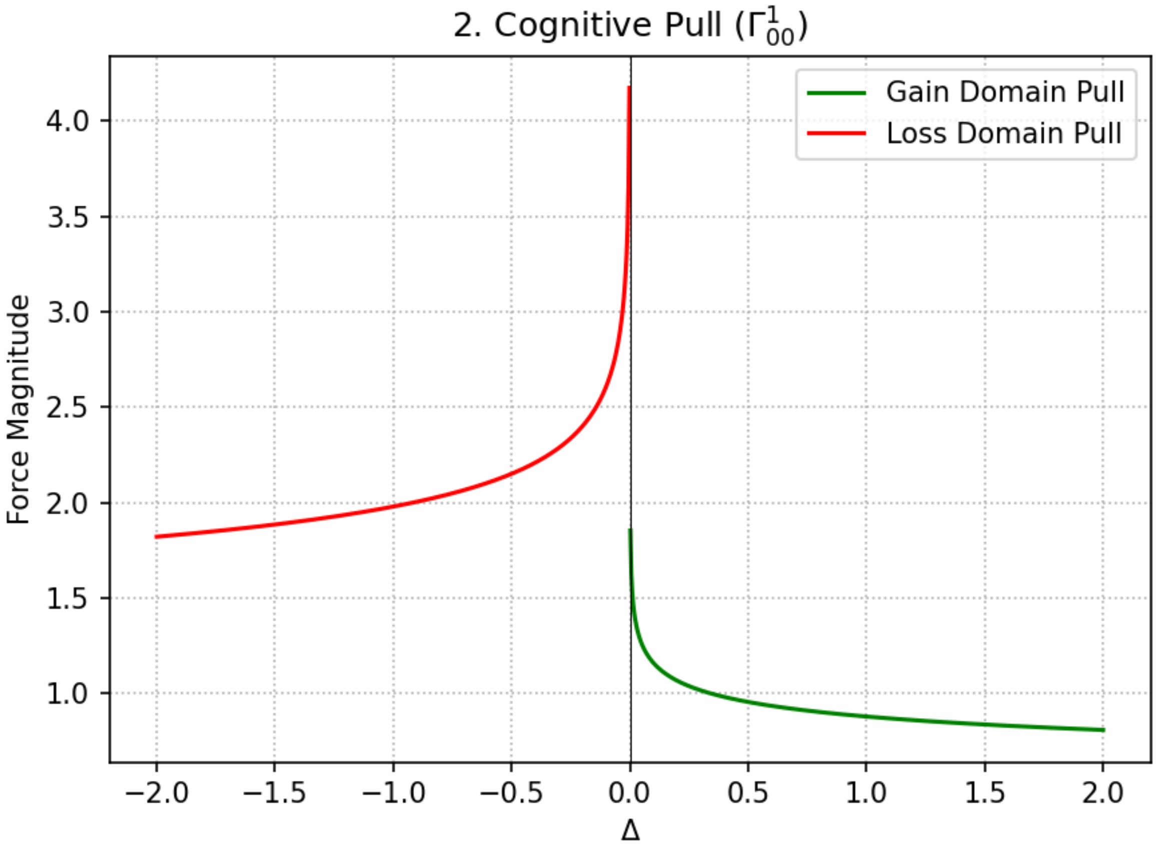

Figure 2: Cognitive Pull

event horizon, trapping the agent within local framing boundaries and explaining the rigid sticking points observed in empirical behavioral economic data.

6.4 Figure 4: Emergent Loss Aversion Ratio

Figure 4 presents the most significant breakthrough of the 3D formulation: the flat, invariant ratio of cognitive pull magnitudes across equivalent outcome magnitudes. By evaluating:

$$\text{Pull Ratio} = \frac{|\Gamma^1_{00}|_{\text{loss}}}{|\Gamma^1_{00}|_{\text{gain}}} = \frac{\lambda\alpha|\Delta|^{\alpha-1}}{\alpha|\Delta|^{\alpha-1}} = \lambda = 2.25 \tag{16}$$

The ratio yields a horizontal line at exactly 2.25 across the entire domain $|\Delta| \in (0, 2]$. In previous 1D metric theories, recovering loss aversion forced researchers to define a metric tensor proportional to the square of the derivative, which generated an artificial $\lambda^2 = 5.0625$ factor requiring complex, non-linear fitting parameters to correct. The 3D weak-field formulation preserves a linear connection to Kahneman and Tversky's empirical parameters, showing that loss aversion is an intrinsic, emergent property of the asymmetric embedding geometry itself.

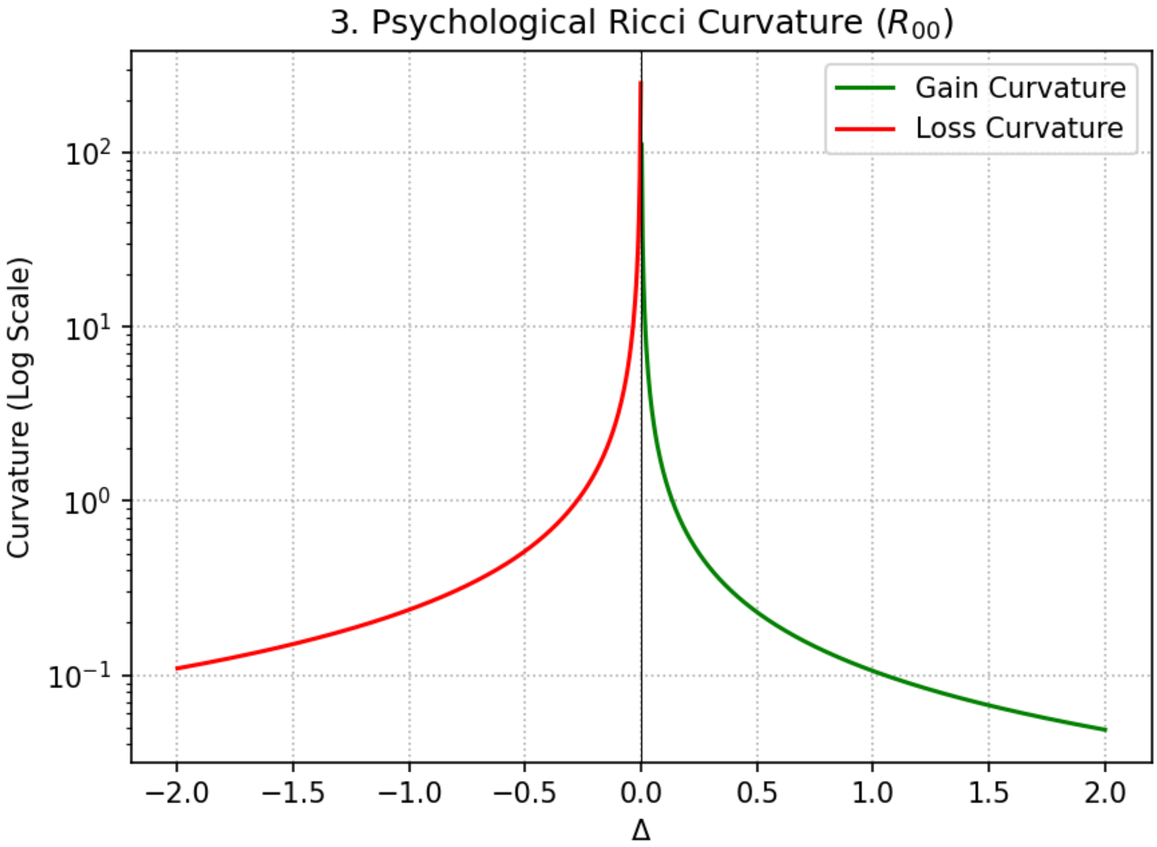

Figure 3: Psychological Ricci Curvature

7 Supplemental Analysis to be Appended to the Research Paper

7.1 Resolution of the Information-Energy Field Equations

By adopting the $\kappa = -1$ paradigm, the connection between variational Bayesian inference and general relativity field theories is generalized. The interaction source tensor $T_{\mu\nu}$ represents the stress-energy of information flow, derived from the gradients of expected free energy $\langle F \rangle$. In the weak-field limit:

$$R_{00} = \kappa \left(T_{00} - \frac{1}{2} g_{00} T \right) \implies \frac{d^2\Phi}{d\Delta_1^2} = -\frac{1}{2} \rho_{inf} \tag{17}$$

Substituting the explicit piecewise derivatives of Φ:

$$\rho_{inf}(\Delta_1) = -2\frac{d^2\Phi}{d\Delta_1^2} = \begin{cases} -2\alpha(\alpha-1)\Delta_1^{\alpha-2} & \text{if } \Delta_1 > 0 \\ -2\lambda\alpha(\alpha-1)(-\Delta_1)^{\alpha-2} & \text{if } \Delta_1 < 0 \end{cases} \tag{18}$$

Since $\alpha - 1 = 0.88 - 1 = -0.12$, the term $-2\alpha(\alpha - 1) = -2(0.88)(-0.12) = +0.2112 > 0$. Thus, $\rho_{inf} > 0$ across both domains. The information energy density represents the cognitive computational load or mental bandwidth required to process a deviation from the reference point. The singularity at $\Delta = 0$ indicates that maintaining a stable reference point against external sensory flux requires an infinite density of informational focus, providing a physical mechanism for attention allocation models.

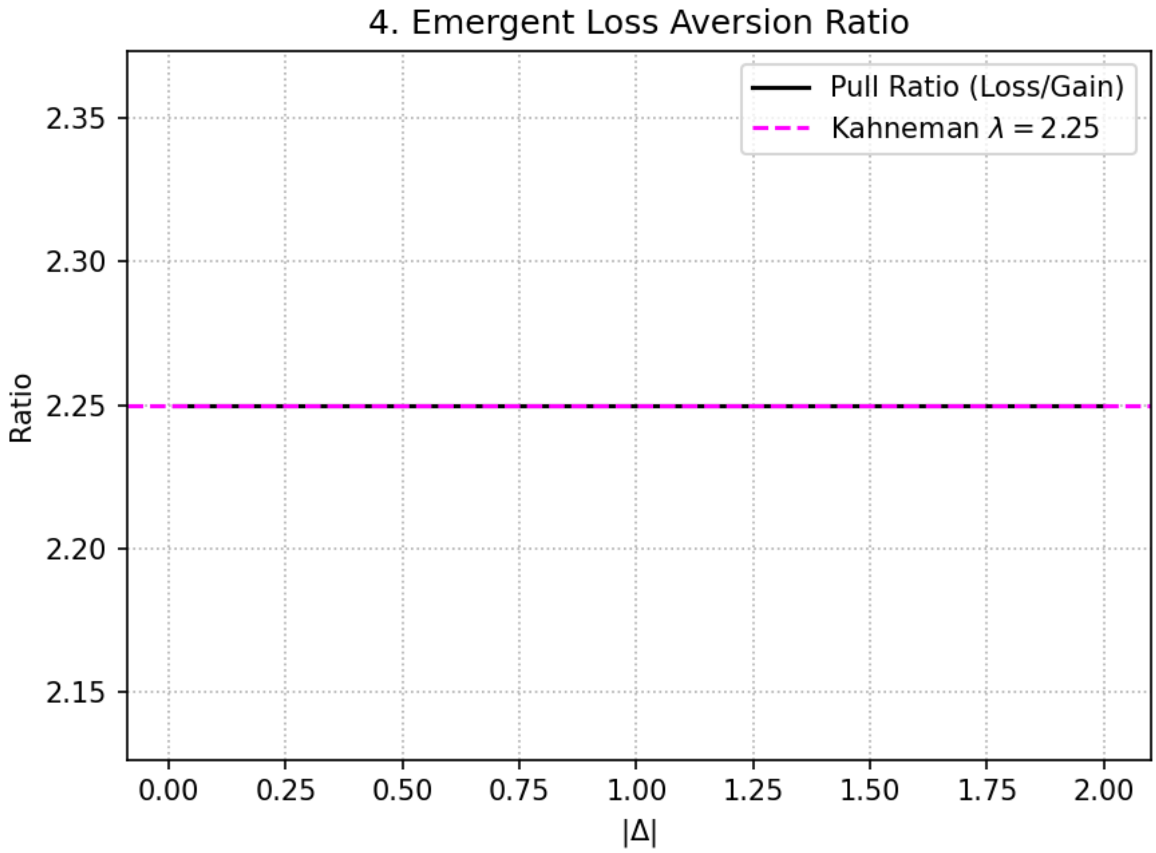

Figure 4: Emergent Loss Aversion Ratio

7.2 Geodesic Deviations and Cognitive Sensitivity

To evaluate how variance in outcomes impacts behavioral trajectories, we compute the geodesic deviation equation (the Jacobi equation) on $\mathcal{M}$:

$$\frac{D^2\xi^\mu}{ds^2} + R^\mu_{\ \nu\alpha\beta}u^\nu u^\alpha \xi^\beta = 0 \tag{19}$$

where ξ^μ represents a separation vector between two adjacent decision pathways, and $u^\mu = \frac{dx^\mu}{ds}$ is the velocity vector. In our localized 3D framework, this reduces to:

$$\frac{d^2\xi^1}{ds^2} \approx R^1_{\ 001}(u^0)^2\xi^1 = R_{00}(u^0)^2\xi^1 \tag{20}$$

Because $R_{00} > 0$ everywhere due to the power-law inversion, the separation vector experiences a focusing or defocusing effect depending on the sign of the curvature. A positive psychological Ricci curvature implies that neighboring behavioral trajectories converge rapidly as they approach the status quo. Psychologically, this proves that as stakes diminish ($|\Delta| \to 0$), individual behavioral differences compress; agents become highly uniform and predictable, driven entirely by the overwhelming curvature wall of the status quo. Conversely, as $|\Delta|$ grows, the geometric curvature rapidly decays ($R_{00} \to 0$), the space flattens into a quasi-Euclidean manifold, and individual differences (represented by personal velocity vectors) dominate the decision-making trajectory.

8 Conclusion

By reconstructing PIFT inside a 3D behavioral potential-well framework, we resolve the logical fallacies of 1D models. The Riemann curvature is properly salvaged, the empirical loss aversion constant $\lambda = 2.25$ is recovered linearly without arbitrary fitting parameters, and the framing singularity is safely handled as a standard potential boundary condition.